# Feminine deodorants: HMTc Standard

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CITE AS

Pendergrass K. "Feminine deodorants: HMTc Standard". Heavy Metal Certified, 2026. https://heavymetalcertified.com/standards/feminine--deodorants. CC BY 4.0.

CORRESPONDENCE

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KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

## Abstract

The Heavy Metal Tested & Certified heavy-metal-safety standard for feminine deodorants, leave-on and rinse-off (Feminine Care), evaluated on the finished product as placed on market at the sovereign mucosal-contact cosmetic tier. Lead reads across the German BVL 2017 toothpaste tier (500 µg/kg); cadmium, total arsenic and total mercury bind the strictest general-cosmetic value and nickel the only sovereign cosmetic nickel limit; the remaining analytes are mandatory controls or substantiated not-material determinations.

## 1 | How every number on this standard is set

HMTc Method v2.0 (government-limit default). Device forms carry no sovereign heavy-metal content limit, so every panel cell resolves to a mandatory analytical control or a substantiated not-material determination; no occurrence percentile is computed and no number is invented. Cosmetic forms read across the strictest sovereign mucosal-contact cosmetic contaminant limit. Two basis families (device content, cosmetic content) are never conflated, and neither is a leachate/migration number. Provenance vocabulary: {regulatory-alignment, read-across, not-material, control, ratchet}.

## 2 | Master limit table

All values in ppb (µg/kg), as_sold basis. A brand meeting every value qualifies for the mark.

| ANALYTE | TIER | HMTC LIMIT | BASIS |
|---|---|---|---|
| Lead (Pb) | Tier 1 | 500 | human-grade read-across (DE) |
| Cadmium (Cd) | Tier 1 | 100 | government maximum level |

| | | | |
|---|---|---|---|
| Total arsenic (tAs) | Tier 2 | 500 | government maximum level |
| Inorganic arsenic (iAs) | Tier 1 | ≤ tAs | bounded by tAs (reflex speciation) |
| Mercury, total (tHg) | Tier 1 | 100 | government maximum level |
| Methylmercury (MeHg) | Tier 1 | ≤ tHg | bounded by tHg (reflex speciation) |
| Nickel (Ni) | Tier 2 | 10,000 | government maximum level |
| Tin (Sn) | Tier 2 | 3,750,000 | human-grade read-across (European Union) |
| Aluminium (Al) | Tier 2 | 560,000 | human-grade read-across (European Union) |
| Total chromium (Cr) | Tier 2 | 50 | human-grade read-across (AU) |
| Hexavalent chromium (Cr-VI) | Tier 2 | 5 | human-grade read-across (European Union) |

## 3 | Derivation — where each number comes from

Every cell traces to one named sovereign maximum level (the strictest for the nearest analog), or to an explicit control, not-material, or species-bounded determination. No occurrence statistic selects a certification value.

| ANALYTE | HMTC | SOVEREIGN ANALOG / DETERMINATION |
|---|---|---|
| Lead (Pb) | 500 | DE — human-food analog |
| Cadmium (Cd) | 100 | government maximum level |
| Total arsenic (tAs) | 500 | government maximum level |
| Inorganic arsenic (iAs) | ≤ tAs | iAs carries no product-specific sovereign cell; it is a component of and bounded by its parent total tAs (iAs <= tAs), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Mercury, total (tHg) | 100 | government maximum level |

| | | |
|---|---|---|
| Methylmercury (MeHg) | ≤ tHg | MeHg carries no product-specific sovereign cell; it is a component of and bounded by its parent total tHg (MeHg <= tHg), an enforceable ceiling via reflex speciation rather than an empty cell. |
| Nickel (Ni) | 10,000 | government maximum level |
| Tin (Sn) | 3,750,000 | European Union — human-food analog |
| Aluminium (Al) | 560,000 | European Union — human-food analog |
| Total chromium (Cr) | 50 | AU — human-food analog |
| Hexavalent chromium (Cr-VI) | 5 | European Union — human-food analog |

## 4 | Cite this standard

Pendergrass, K. "Feminine deodorants." In Heavy Metal Tested & Certified (HMTc) Feminine Care standard, version 1.0 (2026-08-14). Institute of Contaminant Standards (ICS), 2026.

## References


[1] AU — human-food maximum levels
Binds Cr. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[2] DE — human-food maximum levels
Binds Pb. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL

[3] European Union — human-food maximum levels
Binds Sn, Al, Cr-VI. Strictest applicable human-food sovereign maximum for the nearest analog.
MAXIMUM LEVEL



### STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this reference. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

### COMPETING INTERESTS

This briefing is published by the Heavy Metal Tested & Certified (HMTc) program, operated by the Institute of Contaminant Standards. Its statements about government regulations, toxicology, and occurrence are sourced to the cited primary and expert-body records and to the independently maintained Heavy Metal Index; the program's own certification thresholds are not used as evidence for any factual claim made here.

### DATA AVAILABILITY

The works this reference draws on are listed in full in the references above and on the

### PUBLICATION HISTORY

briefing page at https://heavymetalcertified.com/standards/-feminine-deodorants. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/standards/-feminine-deodorants.